HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Designed and implemented **Python** and **Bash** automation scripts to streamline research code packaging and deployment processes, reducing manual effort by an estimated **40%**.
- Developed and maintained CI/CD pipelines using **Jenkins** and **Git**, enabling faster iteration cycles for research codebases and improving deployment reliability.
- Built reproducible polyglot environments using **Docker** and **Linux** scripting, supporting rapid research iteration across **Python** and native extensions.
- Collaborated with research teams to establish and enforce infrastructure best practices, advising on scalable and reliable research workflows.
- Strengthened packaging standards for large research codebases, incorporating modern tooling like **uv** and **hatchling** for improved dependency management.
- Partnered with cloud infrastructure teams to scale research workloads on **AWS**, enhancing execution efficiency and resource utilization.
- Created solutions for researchers and developers to profile and monitor application performance, gaining insights into infrastructure bottlenecks.

SKILLS

DevOps & Infrastructure – CI/CD, Jenkins, Git, Docker, Kubernetes, Nix, Conda, Pixi, uv, hatchling, Linux internals, software deployment, package management

Programming Languages – Python, Bash, CUDA

Cloud Platforms – AWS

Scripting & Automation – Python scripting, Shell scripting

System Monitoring & Profiling – Application performance monitoring, Infrastructure performance monitoring

PROJECTS

Scalable Research Environment Deployment (HCL Technologies)

- Engineered a containerized environment using **Docker** and **Kubernetes** to support diverse research workloads, enabling consistent execution across development and production.
- Developed **Python** scripts for automated environment setup and dependency management, facilitating rapid onboarding of new researchers and projects.
- Integrated custom build tooling with **Jenkins** CI/CD pipelines to automate the testing and deployment of research artifacts, reducing manual intervention by approximately **30%**.

Automated Code Packaging for Research (HCL Technologies)

- Designed and implemented a standardized code packaging system using **Python** and modern tooling like **uv**, ensuring reproducible builds for large research codebases.
- Created automation scripts to manage dependencies and build artifacts, improving the reliability and speed of code deployment for quantitative research teams.
- Documented best practices for code packaging and version control, empowering research teams to manage their code more effectively.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)